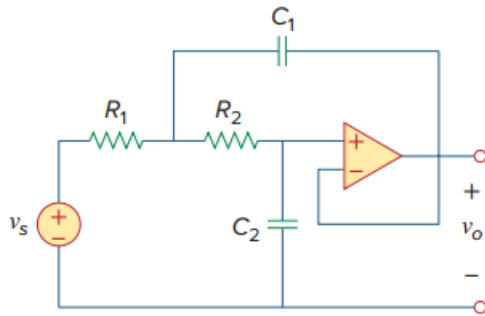


Problem

Using Fig. 8.109, design a problem to help other students better understand second-order op amp circuits.

Figure 8.109:

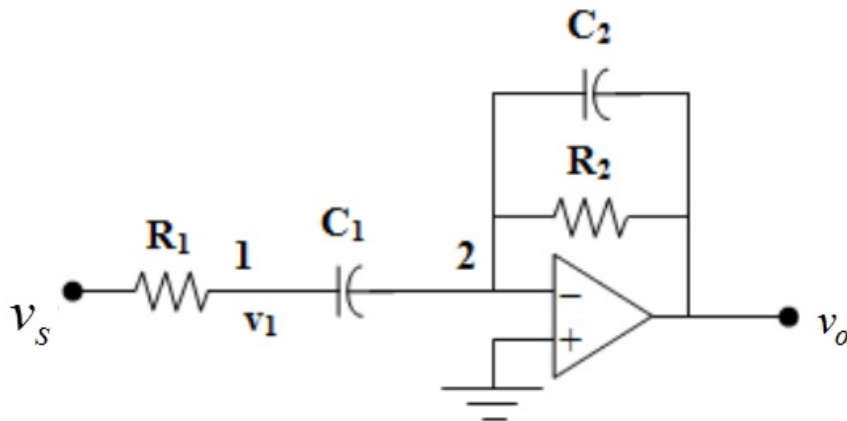


Step-by-step solution

Step 1 of 3

Refer to Figure 8.109 in the text book.

Identify the nodes and draw the equivalent circuit.



Step 2 of 3

The current passing through the resistor R_2 is,

$$i_{R2} = \frac{v_o}{R_2}$$

The current passing through the capacitor C_2 is,

$$i_{C2} = C_2 \frac{dv_o}{dt}$$

The current enters at the inverting terminal of the op amp is,

$$\begin{aligned} i &= i_{R2} + i_{C2} \\ &= \frac{v_o}{R_2} + C_2 \frac{dv_o}{dt} \end{aligned}$$

Step 3 of 3

Use Kirchhoff's voltage law to write the expression for v_s .

$$v_s = iR_1 + \frac{1}{C_1} \int i dt$$

Substitute $\frac{v_o}{R_2} + C_2 \frac{dv_o}{dt}$ for i .

$$\begin{aligned} v_s &= \left(\frac{v_o}{R_2} + C_2 \frac{dv_o}{dt} \right) R_1 + \frac{1}{C_1} \int \left(\frac{v_o}{R_2} + C_2 \frac{dv_o}{dt} \right) dt \\ &= \frac{R_1}{R_2} v_o + R_1 C_2 \frac{dv_o}{dt} + \frac{1}{R_2 C_1} \int v_o dt + \frac{C_2}{C_1} v_o \\ &= R_1 C_2 \frac{dv_o}{dt} + \left(\frac{R_1}{R_2} + \frac{C_2}{C_1} \right) v_o + \frac{1}{R_2 C_1} \int v_o dt \end{aligned}$$

Differentiate on both the sides.

$$\begin{aligned} R_1 C_2 \frac{d^2 v_o}{dt^2} + \left(\frac{R_1}{R_2} + \frac{C_2}{C_1} \right) \frac{dv_o}{dt} + \frac{1}{R_2 C_1} v_o &= \frac{dv_s}{dt} \\ \frac{d^2 v_o}{dt^2} + \left(\frac{1}{R_2 C_2} + \frac{1}{R_1 C_1} \right) \frac{dv_o}{dt} + \frac{1}{R_1 R_2 C_1 C_2} v_o &= \frac{1}{R_1 C_2} \frac{dv_s}{dt} \end{aligned}$$

Thus, the second order differential equation of the circuit is

$$\boxed{\frac{d^2 v_o}{dt^2} + \left(\frac{1}{R_2 C_2} + \frac{1}{R_1 C_1} \right) \frac{dv_o}{dt} + \frac{1}{R_1 R_2 C_1 C_2} v_o = \frac{1}{R_1 C_2} \frac{dv_s}{dt}}$$